

Hitesh Rana

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Experience

Mercari

Bengaluru, India

Software Engineer - 2

Dec 2025 - Present

- Worked on the production **Rule Engine** for fraud and risk decisions, handling peak traffic of **10K payments/min** and **1.5M rule evaluations/min** with up to **150 rules per transaction**.
- Implemented dynamic JSONLogic rule evaluation with typed datasets, on-demand feature fetching, dry-run execution, allow/block overrides, and explainable decisions.
- Expanded the Rule Engine from payments to multi-tenant item-risk detection, supporting **1K listings/min** and up to **2M category-filtered rule evaluations/min**.
- Built **jsonlogic2sql**, a dialect-aware JSONLogic-to-SQL library now used in production for BigQuery-based historical rule simulation, **reducing pre-release validation effort by 80%**.
- Built scheduled Dataflow ETL and BigQuery jobs for risk data pipelines, supporting online decisions and offline analytics alongside Flink and Cloud Spanner.
- Delivered **Rule Analytics** end to end, giving anti-fraud operations real-time visibility into performance across **200+ active rules** to guide operational decisions.
- Drove adoption of POS terminal data in the warehouse as a single source of truth, **eliminating duplicate data handling across 5+ services** and improving downstream consistency.

Mercari

Tokyo, Japan

Software Engineer - 1

Oct 2024 - Nov 2025

- Architected high-volume asynchronous order-cancellation processing, **cutting latency by 40%**, preventing duplicate operations, and saving **¥7.4M/month** through improved fraud detection.
- Refactored and optimized a **very high RPS microservice** by introducing an efficient caching layer, **cutting database query load by 85%** and **saving \$40K/month** in cloud infrastructure costs.
- Owned full lifecycle development of an internal microservice for real-time abuse risk scoring, maintaining **p99 latency under 100 ms** and automating investigation workflows, **saving analysts up to 5 minutes per case**.
- Streamlined Flink deployments by consolidating and decommissioning anti-fraud rules, **reducing infrastructure costs by 30%** while improving reliability.

Mercari

Tokyo, Japan

Software Engineer Intern

Feb 2024 - Apr 2024

- Migrated legacy Spinnaker pipelines to a **GitOps-based system** (Argo CD, Flux, Tekton), **reducing deployment time by 40%** and accelerating feature delivery across teams.
- Engineered a data synchronization feature **leveraging gRPC for low-latency data exchange**, to update stale task statuses across distributed services, **improving data consistency and reducing process disruptions by 75%**.

Projects

Chronoverse | Go, gRPC, Next.js, Kafka, PostgreSQL, Redis, ClickHouse, Meilisearch, Docker, Kubernetes, Distributed Systems

- Architected a distributed workflow scheduler supporting **10,000+ concurrent executions** with **99.9% reliability**, using durable job leases, idempotency keys, and a PostgreSQL transactional outbox.
- Cut system latency by **40%** using **gRPC** for service communication and **Kafka** for asynchronous workflow and job processing.
- Engineered a **multi-node Kubernetes deployment** with a runtime agent and secured Docker proxy on each execution node, plus HPA and Kafka-lag-based worker scaling.

Skills

Languages: Go, TypeScript, Python, Java, C/C++

Libraries & Frameworks: Next.js, React.js, FastAPI, Flask

Data Systems: PostgreSQL, Redis, Cloud Spanner, BigQuery, ClickHouse, Meilisearch, MySQL

Systems & Cloud: gRPC, Kafka, Flink, Docker, Kubernetes, GCP, AWS, Datadog, GitHub Actions

Education

Maharaja Surajmal Institute of Technology, New Delhi

B.Tech. in Computer Science and Engineering

CGPA: 9.1

Jun 2020 - Aug 2024